

University of Engineering and Technology, Taxila

Computer Networks Lab



Project Report
SMART PARKING SYSTEM

Submitted to

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SMART PARKING SYSTEM

1. Overview:

The Smart Parking System is an IoT-based solution designed to automate parking management using sensors, network devices, and a central control interface. The system detects vehicle entry/exit, monitors parking slot occupancy, and provides real-time status updates to administrators via a tablet controller.

2. Objectives:

- Automate gate control using motion sensors.
- Detect vehicle presence using metal sensors.
- Display parking slot status on LCDs.
- Enable remote monitoring via a home gateway and tablet.

3. System Architecture:

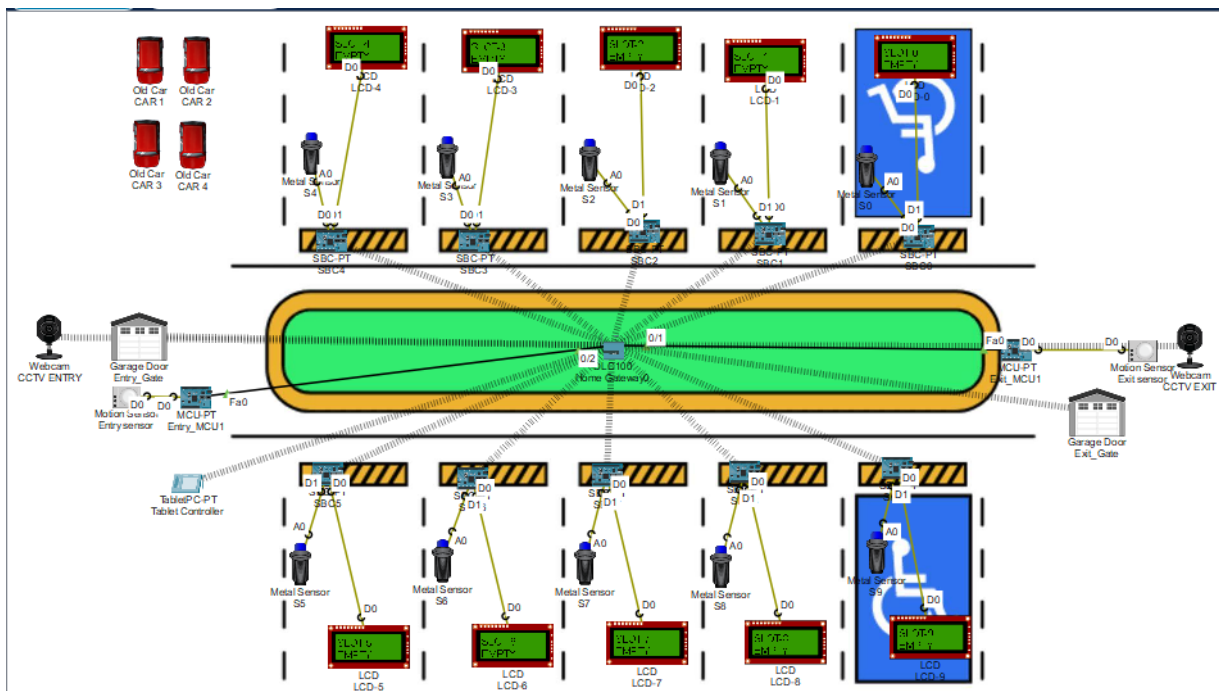
The system consists of the following components:

3.1 Hardware Components (Simulated in CPT):

- Motion Sensors – Trigger gate opening.
- Metal Sensors – Detect parked vehicles.
- LCD Displays – Show “Empty” or “Occupied.”
- CCTV Camera – Monitors entry/exit.
- Home Gateway – Connects all devices to the network.
- Tablet Controller – Displays real-time status of all devices.
- Switches (S1–S4) – Connect sensors and devices locally.
- PC and Laptop – For configuration and monitoring

3.2 Network Topology:

- LAN 1: Entry/Exit Gate (Motion Sensor, CCTV, LCD)
- LAN 2: Parking Slot Sensors (Metal Sensors, LCDs)
- Gateway: Home Gateway (Router) connects all subnets.
- Wireless Network: Tablet connects via Wi-Fi to gateway.



4. Implementation in Cisco Packet Tracer:

4.1 Device Configuration:

- Sensors: Connected to switches via GPIO pins.
- LCDs: Programmed using IoT logic to display slot status.
- CCTV: Streams to PC and tablet via IP camera module.
- Home Gateway: Configured with DHCP and VLANs for segmentation.
- Tablet: Runs a dashboard app showing sensor states.

4.2 Network Configuration:

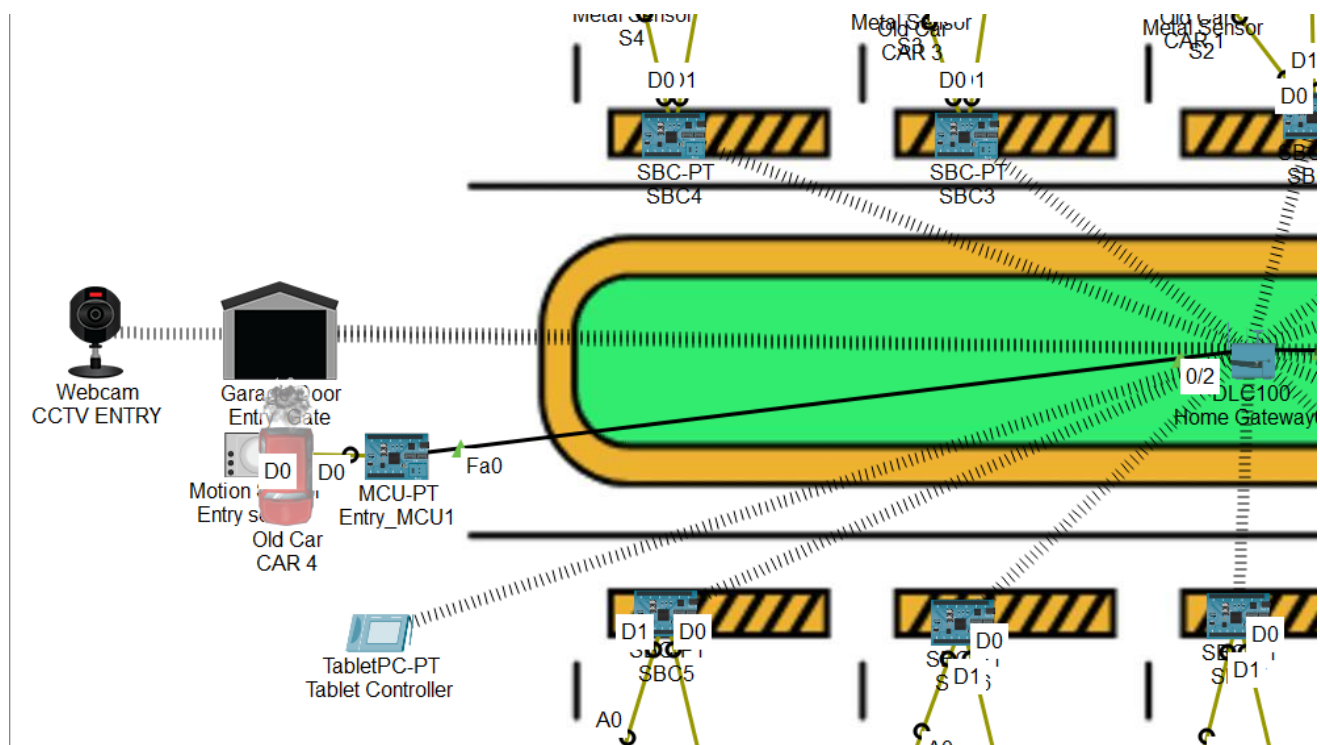
- IP Addressing:
 - Home gateway: 192.168.25.1/24
 - MCU1 Network: 192.168.25.128/24
 - MCU2 Network: 192.168.25.129/24
 - DHCP Automatic IP addressing in tablet and all IOT devices connect to Home gateway through Internet

4.3 IoT Programming:

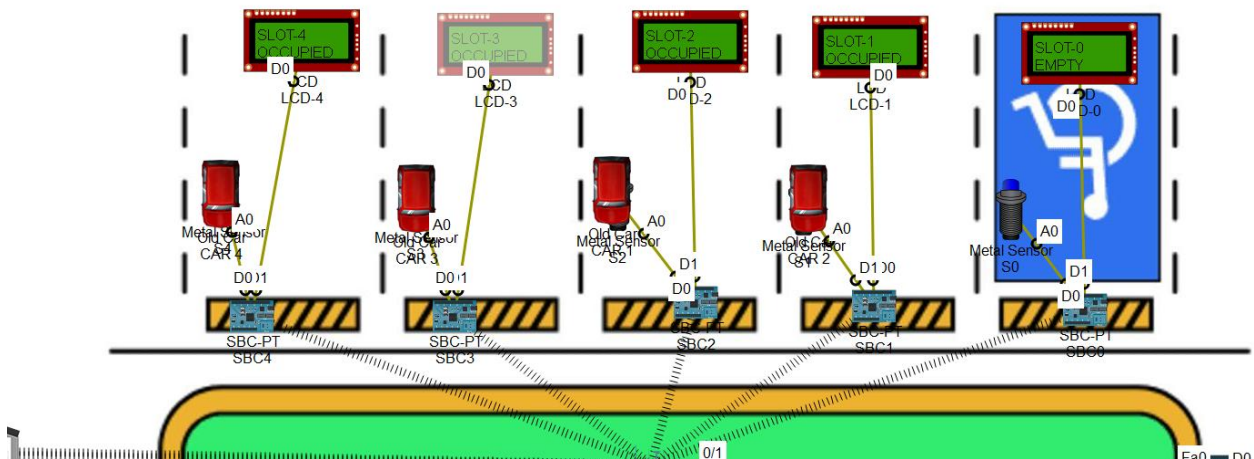
- Motion sensor → GPIO → Gate actuator.
- Metal sensor → GPIO → LCD update.
- Tablet apps read and display messages and show all IOT devices connected to sensors via gateway.

5. How the System Works:

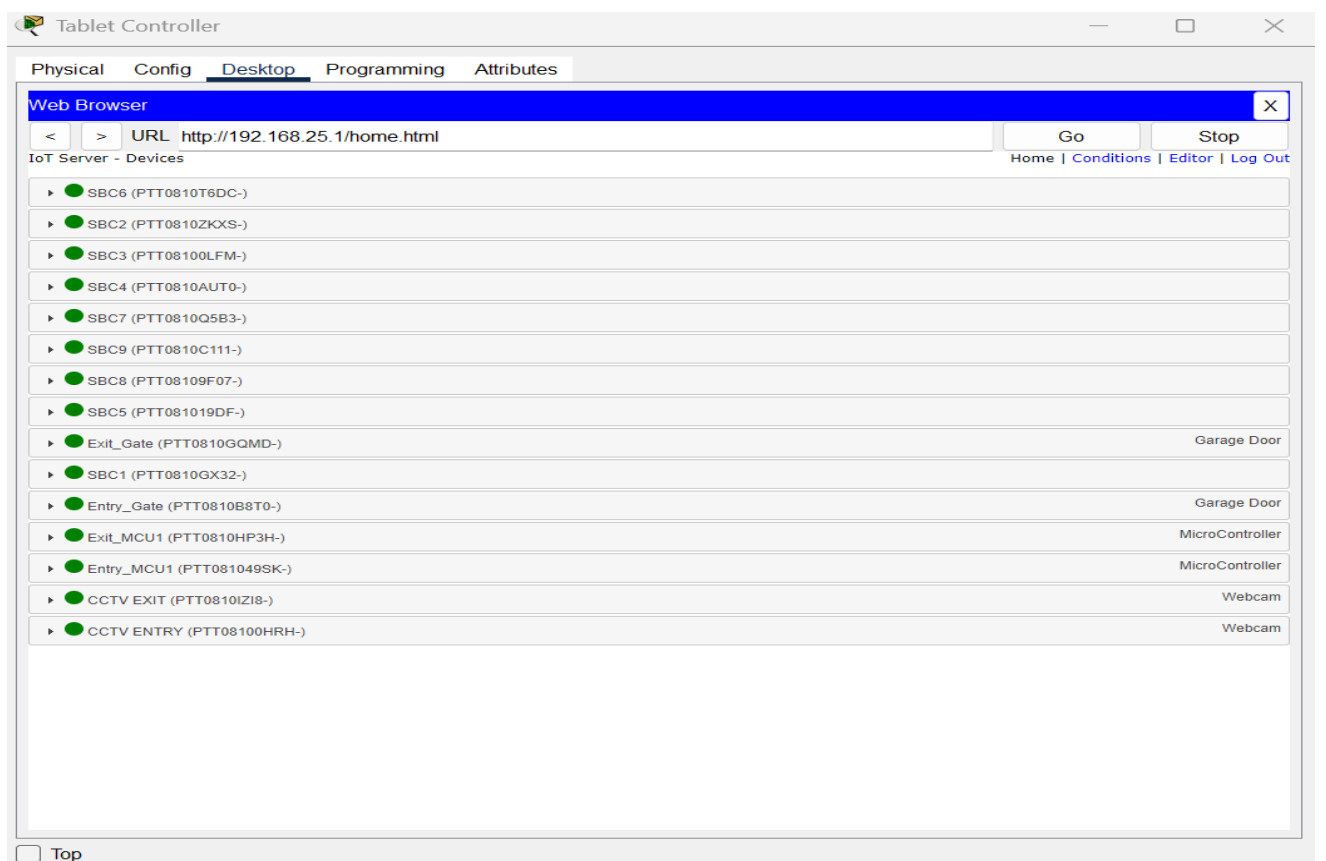
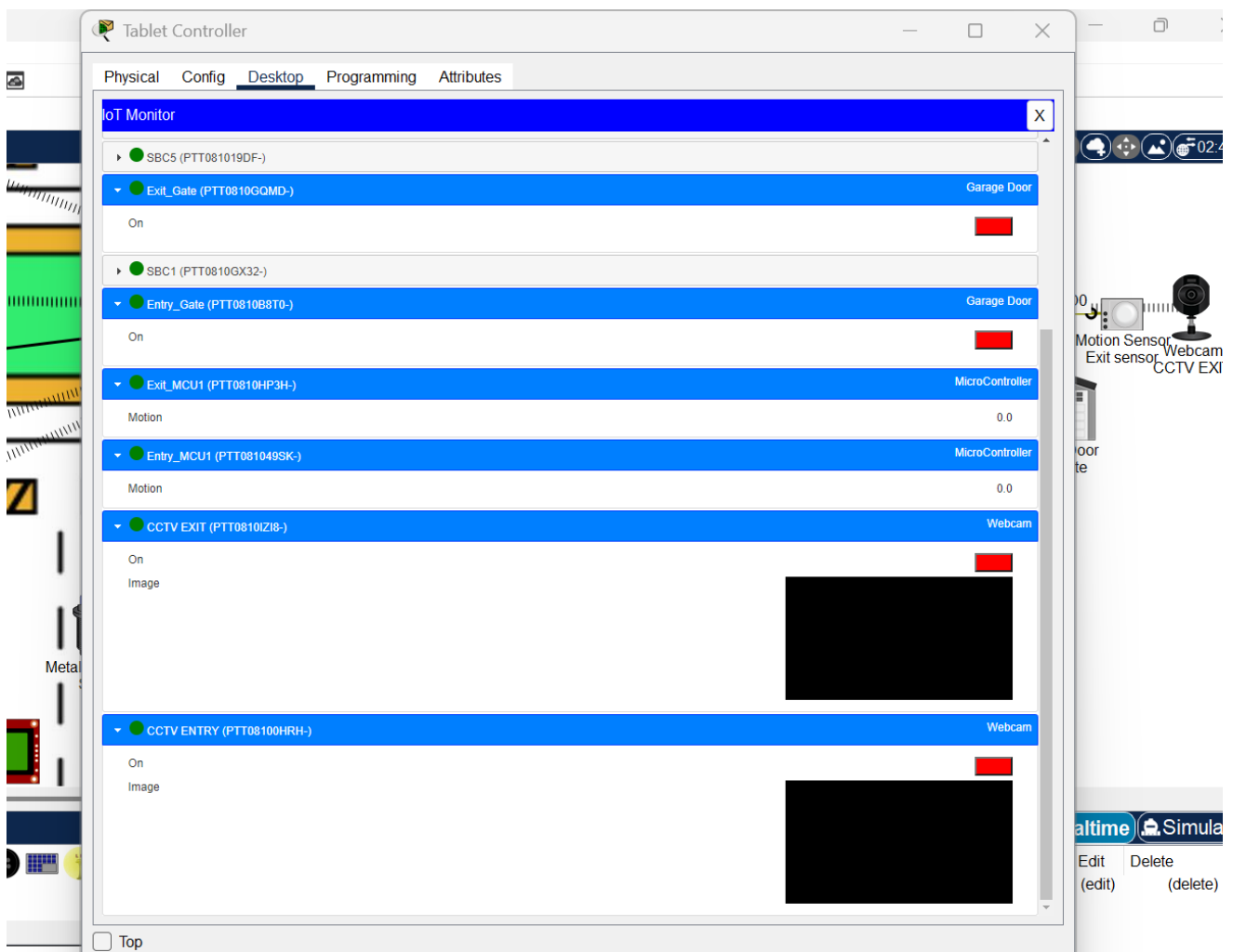
- Entry: Motion sensor detects vehicle → gate opens → CCTV logs entry.

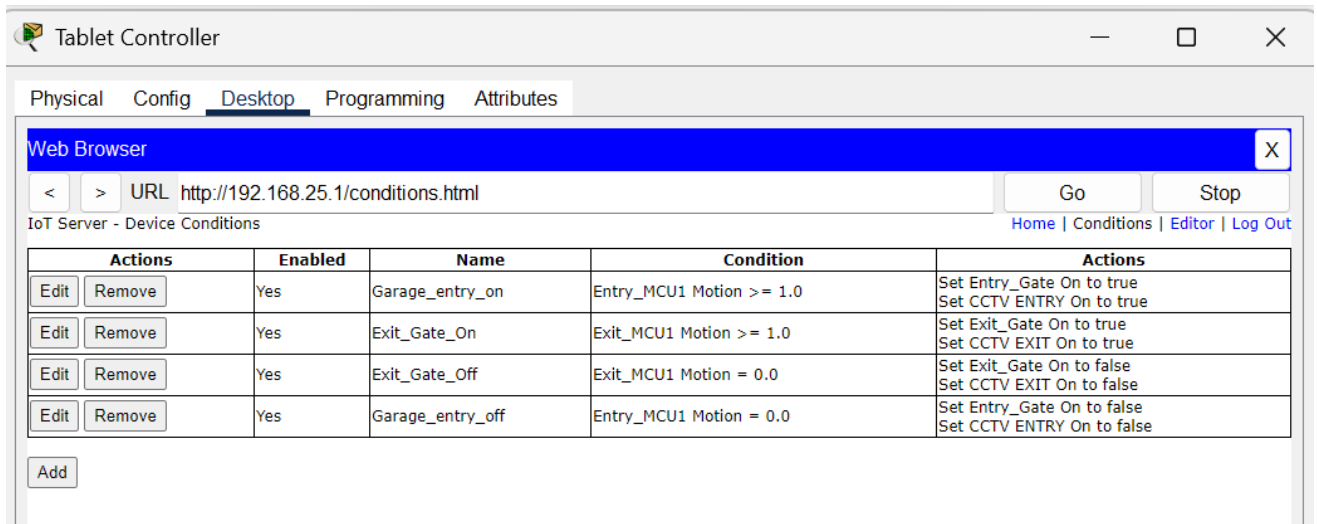


- Parking: Metal sensor detects car in slot → LCD updates to “Occupied.”



- Monitoring: Tablet shows real-time map of IOT devices.





- Exit: Motion sensor triggers gate opening → CCTV “ON”

